Workbook statistics for medical engineers

Statistics for Medical Engineering (Kungliga Tekniska Högskolan)



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KTH Royal Institute of Technology



Workbook

CM2018 HT24
Statistics for Medical Engineering

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Types of Data:

◊ Interval: Numerical, often (not necessarily) continuous.



◊ Ordinal: Ordered categories.
Data can be organized. Categories can be ranked.
However, differences between ranks do not have a mathematical meaning. This means the intervals between the data points are not necessarily equal



◊ Nominal: Individual categories, no order.
Most basic level of measurement. Data can be categorized.
Not possible to rank the categories.



Ordinal and nominal data are called categorical data.

Analyzing interval, ordinal and nominal data

Data type	Description	Analysis
Interval, normal non-normal	Mean, SD…. Median, quartiles	t-test, ANOVA… non-parametric
Ordinal	Median, quartiles	Non-parametric,…
Nominal	Frequency / contingency table	χ^2test, Fisher exact test

Great website to learn more about the types of data: https://www.mygreatlearning.com/blog/types-of-data/

Nominal data
- t-test, anova
- Bar chart, Pie chart

Ordinal data
- Kruskal-Wallis-Test, Mann-Whitney U test
- Line diagram, Box plot

Metric
- Chi-square-test
- Scatter diagram, histogram

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Distributions:

Binomial Distribution:

The **binomial distribution** is a discrete probability distribution that models the number of successes in a fixed number of independent trials of a binary (success/failure) experiment. One of the most commonly used probability distributions in statistics.

Applications:
Quality Control: Counting defective items in a batch.
Medicine: Number of patients recovering after a treatment in a sample group.
Marketing: Number of costumers responding to a promotion.

If each trial has probability of success $p=\theta$, how probable is it that *k* out of N trials will succeed?

Example: 20 patients treated with a drug that has a 75% chance of being effective. How probable is that x of that patients will be cured?
x = 1:20; y = dbinom(x, 20, 0.75); plot(x, y, type = "h", lwd = 4, col = 2

Poisson Distribution:

The **Poisson distribution** is a discrete probability distribution that models the number of events occurring in a fixed interval of time, space, or other continuous domain, under specific conditions. It is commonly used in a situations where events happen randomly and independently at a constant average rate.

Applications:
Traffic and Queues: Counting cars passing a checkpoint in an hour.
Call Centers: Number of calls received in a fixed time.
Biology: Number of mutations in a strand of DNA over a certain length.
Finance: Modeling rare events such as defaults or claims.
Astronomy: Number of stars appearing in a specific region of the sky.

Conditions for using the Poisson Distribution:

1. Events occur randomly in time or space.
2. The average rate (λ) is constant
3. The number of events in disjoint intervals is independent.

If events occur at a typical rate of λ, how probable is it that exactly *k* events will occur?

Example: a radioactive isotope emits photons at a mean rate of 8 photons per second. How probable is it that x photons will be emitted in a second? How probable is it that at least 5 and at most 10 photons will be emitted?
(See slide 12 and 13 in L1-1)

Normal (Gaussian) Distribution:

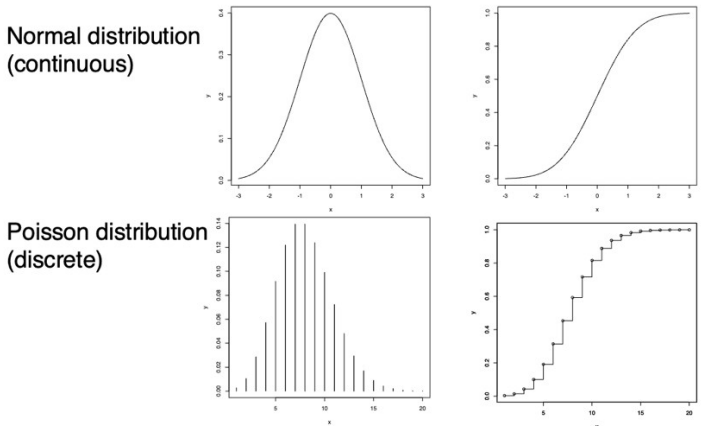
The **normal distribution**, also known as the **Gaussian distribution**, is one of the most important and widely used probability distributions in statistics. It describes data that cluster around a central mean, and its characteristic bell-shaped curve makes is a fundamental concept in probability theory and statistics.

The variable x has values on the real line close to μ, but spread out, with a typical distance from μ of σ.

Examples: Body length, blood pressure,…

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Density function vs. distribution function:



Median: the value that has the same number of observations above it and below it.

Central Limit Theorem:

With increasing sample size, the distribution of the sample mean will approach a normal distribution regardless of the original distribution.

Tests:

T-test:

The t-test is a test statistic computed from the difference between the groups. It assumes that both groups are normally distributed.

Test for normality: qqplot (the distribution must follow the line), density plot, Shapiro-Wilk test (if p-value > 0.05, the data is normally distributed). Normality can be assumed for large datasets.
Shapiro.test(sample_data); qqplot(sample_data); hist(sample_data)

- ◊ **Unpaired t-test:** Comparison between two groups of independent cases.
  Example: Suppose researchers are investigating the effectiveness of a new drug in reducing blood pressure. They have a group of patients who were treated with a new drug (Group A) and another group of patients who received a placebo (Group B).
  t.test(group_A, group_B)

- ◊ **One-sample t-test:** Comparison between a mean of a sample with a known reference mean..
  Example: A group of researchers is investigating the average blood sugar levels in a population to determine if they differ from the standard normal level of 100 mg/dL. They collect blood sugar data from a sample of patients.
  t.test(blood_sugar_levels, mu = 100)

- ◊ **Paired t-test:** Comparison between different observations on the same case. Often a change over time. Assume that the difference are normally distributed (but not that each variable is).

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Example: You are testing a new drug to see if it has an effect on blood pressure. You measure the blood pressure of the same group of individuals before they take the drug (baseline) and then after they have taken the drug for a certain period. You want to determine whether the drug has a statistically significant effect on blood pressure.
t.test(before_treatment, after_treatment, paired = TRUE)

Wilcoxon test:

The Wilcoxon test is a non-parametric alternative to the t-test. It does not require the normality of the distribution.

◊ **Unpaired Wilcoxon test:** does a rank transformation of the values and tests on the sum of ranks in each group. Also called *Mann-Whitney U test*.
- alternative = "greater" → This option is used when you want to test if the first group is less than the second group. In other words, it checks if there is a significant difference with the first group having higher levels.
- alternative = "less" → This option is used when you want to test if the first group is less than the second group. It checks if there is a significant difference with the first group having lower values.
Example: Researchers are conducting a study to compare the pain relief effectiveness of two different pain medications. They have two groups if patients: Group A, treated with Medication X, and Group B, treated with Medication Y. each patient's pain level is measured before and after treatment. The researchers want to determine if there is a significant difference in the pain relief achieved by the two medications.
wilcox.test(group_medication_A_after, group_medication_B_after, alternative = "…")

◊ **Paired Wilcoxon test:** computes the ranks of absolute difference, puts a sign in front, then does a test. It requires interval data.
Example: Suppose researchers are studying the impact of a new exercise program on the cardiovascular fitness of a group of individuals. Each participant's cardiovascular fitness level is measured before and after completing the exercise program. The researchers want to determine if there is a significant difference in cardiovascular fitness before and after the exercise program.
wilcox.test(before_exercise, after_exercise, paire = "TRUE")

Sign test:

The sign test in R is a non-parametric statistical test that is used when you have paired data and you want to determine if there is a significant difference between the two related groups. It's particularly useful when the assumptions of a parametric test (such as the t-test) cannot be met. When dealing with ordinal data, where the ranking or order is important but the intervals between values may not be, the sign test can be suitable choice.
Example: In a radiology department, two different types of monitors were compared by showing 30 clinical chest radiography images to a radiologist, who graded the image quality visually. Each image was displayed twice, first on monitor A and then on monitor B, and on the each occasion the radiologist chose a score on a scale from 1 (worst) to 5 (best).
signTest(monitorA, monitorB, paired = TRUE)

ANOVA test:

The One-way analysis of variance (ANOVA) test is used to compare the means of two or more groups to determine if there are statistically significant differences among those groups means. *The observations need to be independent, and the groups must follow a normal distribution.*

- **One-way ANOVA** → used to test for significant differences in means among three or more independent (unrelated) groups.
- **Two-way ANOVA** → extension of one-way ANOVA, used when you have two independent categorical variables (factors) and you want to analyze their individual effects and the interaction effect on the dependent variable.

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- **Multivariate ANOVA →** MANOVA is used when you have two or more dependent variables and you want to determine whether there are statistically significant differences among groups with respect to theses dependent variables.
- **Covariance ANOVA →** ANCOVA combines aspects of ANOVA and regression. It is used when you want to compare group means while controlling for the influence of a covariate (a continuous variable).

Example: Imagine a clinical trial where a pharmaceutical company is testing the efficacy of three different drug treatments (A, B and C) for managing blood glucose levels in diabetic patients. The study involves measuring blood glucose levels after the completion of the treatment period for each patient. The researchers want to determine if there is a significant difference in the mean blood glucose levels among the three drug treatments.

```
Anova_model = aov(blood_glucose ~ ftrug, data = data)
Anova(anova_model)
```

Kruskal-Wallis test:

The Kruskal-Wallis test is a non parametric alternative to the ANOVA test.

Example: Consider a study where researchers are investigating the effect of different exercise regimens on pain reductions in patients with chronic back pain. Patients are randomly assigned to one of three exercise groups. Group A practices yoga, group B follows stretching routine, and group C engages in aerobic exercise. After a defined period, each patient rates their pain level on a scale from 1 to 10.

```
Kruskal.test(pain_reduction ~ exercise, data = data)
```

Friedman test:

The Friedman test is a non-parametric statistical test used to determine whether there is statistically significant difference between the means of three or more paired groups. It is an extension of the Wilcoxon signed-rank test for more than two related samples. The test is often used when the data is not normally distributed.

Example: Imagine a clinical trial testing the effectiveness of a new medication on blood pressure. The trial includes 30 patients, each of whom has their blood pressure measured at three different time: before treatment (baseline), one month after starting treatment, and three months after starting treatment. The goal is to determine if there is a statistically significant change in blood pressure over these three time points.

```
Fried_test = friedman.test(baseline, one_month, three_month, data = data)
```

Post-hoc tests:

Post-hoc tests are statistical tests that are conducted after an initial statistical analysis, typically after an analysis of variance (ANOVA) or a Kruskal-Wallis test, to further investigate and compare specific group differences. Theses tests are used to identify which groups are significantly different from each other when the initial analysis indicates a significant overall difference among groups.

One test is the Tukey *HSD* test to compare groups to each other.

Example: You have three groups: Group A, Group B, and Group C.

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The ANOVA results indicate a significant overall difference among the groups. You are comparing the test scores of students who received three different types of tutoring (Group A, Group B, and Group C) to see if there are significant differences between each group.

```
TukeyHSD(anova_model)
```

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DunnTest (kruskal_model)

Sometimes it is useful to study the interactions between two variables. Interactions refer to the way in which the effect of one variable on a dependent variable depends on the level or value of another variable.

Example: You have three groups: Group A, Group B, and Group C. The ANOVA results indicate a significant overall difference among the groups. You are comparing the test scores of students who received three different types of tutoring (Group A, Group B, and Group C) to see if there are significant differences between each group.

TukeyHSD (anova_model)
DunnTest (kruskal_model)

Sometimes it is useful to study the interactions between two variables. Interactions refer to the way in which the effect of one variable on a dependent variable depends on the level or value of another variable.

Chi-squared test and Fisher test:

The Fisher exact test and the chi-squared test are both statistical tests used to analyze binary or nominal data and assess the association between two binary or nominal variables.

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◊ *Chi-squared test:* a statistical test used to examine the association between two or more binary or nominal variables. It is particularly suitable for analyzing larger datasets. Acceptable if all numbers in the expected table are greater than 5.
Example: Consider a study examining the association between smoking status (smoker or non-smoker) and the occurrence of a respiratory condition (e.g., chronic bronchitis) among a group of patients in a respiratory clinic.

```
data_matrix = matrix(c(smoker_with_condition, non_smoker_with_condition,
smoker_without_condition, non_smoker_without_condition), nrow = 2, byrow = TRUE, dimname =
list(c("Raspiratory Condition", "No Respiratory Condition"), c("Smoker", "Non-Smoker")))
chi_squared_test_result = chisq.test(data_matrix)
```

◊ *Fisher exact test:* is a non-parametric test used to determine if there are non-random associations between two categorical variables in a 2x2 contingency table. It is often employed when the sample sizes are small, and the chi-squared test assumptions may not hold. It is especially useful for situations where the expected cell counts in a contingency table are less than 5 and the chi-squared test may not be appropriate due to its assumptions.
Example: Suppose you are conducting a clinical trial to evaluate the effectiveness of two different treatments (Treatment A and Treatment B) for a rare medical condition. You have 30 patients and want to determine if there is a significant difference in the number of patients who responded to the treatments.

```
data_matrix = matrix(c(10, 5, 5, 10), nrow = 2, byrow = TRUE)
fisher.test(data_matrix)
```

Linear Regression:

Linear Regression is a statistical method to model the relationship between a dependent variable (the outcome you want to predict) and one or more independent variables (predictors) by fitting a linear equation to the observed data. The goal is to find the best fitting line that minimizes the difference between the observed and predicted values. The data needs to be interval.

$$Y = b_0 + b_1 * x$$

Assumptions:
Independence: The n observations are independent of each other.
Normality: The error term has a normal distribution (check with qqplot, Shapiro test).
Linearity: The relationship between x and y is linear (check with residual plots).
Homoscedasticity: The variance of the error term is the same for all x (check with residual plot)

```
residuals(lin_model)
shapiro.test(residuals_cm)
plot(fitted(cube_model), residual_cm, xlab = "fitted", ylab = "Residuals", main = "Residual Plot", las = 1) +
    abline(h = 0, col = "green")
```

Example: how well can we predict blood pressure if we know the age?

```
Lin_model = lm(ft16BP ~ AGE)
```

Example: how well can we predict blood pressure if we know age and sex (and interactions) ?

```
Lin_model = lm(ftBP16 ~ AGE + SEX + AGE:SEX)
Lin_model = lm(ftBP16 ~ AGE*SEX)
```

Key assumptions:

1. **Linearity**: The relationship between the dependent variables and the independent variable is linear.
2. **Independence:** Observations are independent of each other.
3. **Homoscedasticity:** The variance of residuals (errors) is constant across all levels of the independent variables.
4. **Normality:** Residuals are normally distributed.
5. **No Multicollinearity:** Independent variables should not be highly correlated with each other.

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Goodness of fit:

Goodness of fit is a statistical concept used to assess how well a model or statistical analysis fits the observed data. It is a measure of how closely the predicted values form a model or analysis align with the actual observed data points.

How to check goodness of fit:

- R-squared: Represents the portion of the variance that is explained by the model.
  $R^2 = 1 \rightarrow$ perfect model
  $R^2 = 0 \rightarrow$ worthless model
- AIC test: the lower the AIC value the better the model fits the data.
  AIC(lin_model)

Multiple linear regression

Predicting a continuous variable for two other continuous variables.

$$Y = b_0 + b_1 * x_1 + b_2 * x_2 \ldots$$

Example: how well can we predict the systolic blood pressure if we know the age and the diastolic blood pressure?
Lin_regression = lm(SBP16 ~ AGE + ftBP16)

In more complex linear models, we are predicting a continuous variable from one continuous and one categorical variable.

Example: how well can we predict the systolic blood pressure if we know the age and the sex?
Lin_regression = lm(SPB16 ~ AGE + SEX)

Key assumptions:

6. **Linearity**: The relationship between the dependent variables and each independent variable is linear.
7. **Independence:** Observations are independent of each other.
8. **Homoscedasticity:** The variance of residuals is constant across all levels of the predictors.
9. **Normality:** Residuals are normally distributed.
10. **No Multicollinearity:** Independent variables should not be highly correlated with each other.

Polynomial regression:

Polynomial regression is a type of regression analysis used to model the relationship between a dependent variable (target) and one or more independent variables (predictors) when the relationship is not linear but can be better described by a polynomial equation.

$$Y = b_0 + b_1 * x + b_2 * x^2 + \epsilon$$

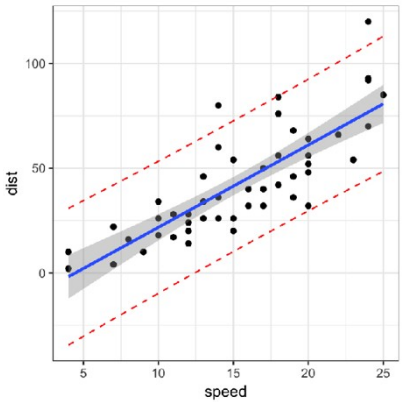
Example: Consider a scenario where researchers are studying the relationship between the dosage of a medication given to patients and their blood pressure levels. The researchers suspect that the relationship may not be linear, and they want to explore whether a polynomial regression model could better capture the complexity of the dose-response relationship.

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Quad_model = lm(BP ~ dose + I(dose^2), data = my_data)

Prediction:

A model is used to predict y from the x's. In R, this can be done with predict() or fitted().



Prediction limits (dashed red lines) are used to assess the uncertainty in making predictions for new data points or individual observations. They take into account the variability of the data, the model's uncertainty, and the error term (ϵ) in the regression equation.

Confidence limits (grey band) estimate the uncertainty associated with the population parameters (i.e., coefficients) in your linear regression model. These intervals provide information about the range within the true coefficients are likely to fall within which the true coefficients are likely to fall with a certain level of confidence.

In general, prediction intervals are much wider than confidence intervals.

Generalized linear models:

Generalized linear models are a class of statistical models that extend linear regression to handle non-nontinuous dependent variables, which may have a non-linear relationship with the predictors.

- *Logistic regression:* is commonly used when the dependent variables is binary ir categorical. It models the probability of a binary outcome (e.g., 0 or 1, „yes" or „no") as a function of one or more independent variables. It is recommended to have a number of observations equal to 10 times the number of independent variables.
  Example: Suppose you are a medical researcher working on a project to predict the likelihood of a patient having a heart attack based on certain medical and lifestyle factors
  model = glm(hear_attack ~ lifestyle, family = „binomial", data = mydata)

- *Ordinal logistic regression:* Ordinal logistic regression, also known as ordinal regression or ordinal logit, is a statistical modeling technique used when the dependent variable is ordinal in nature. The paralell regression needs to be satisfied (*Brant test*). Probability needs to be higher than 0.05.
  Example: you want to investigate how various patient charectaristics, such as age, comorbidities, abd blood test results, affect the severity of illness in patients admitted to a hospital. The severity of illness is catagorized into three levels: „Mild", „Moderate", and „Severe".
  model = polr(illness ~ age + blood_test, data = mydata, Hess = TRUE)

- *Poisson regression:* Poisson regression is used to model data where the dependent variable represents the number of occurences of an event in a fixed unit of time, space, or some other interval (counts). The poisson regression model is particurlarly suited for data where the counts are non-negative integers and have no upper limit.
  Example: Suppose you are a medical researcher studying hospital readmissions of patients with particular chronic condition, such as diabetes. You want to understand how various patient characteristics and hsopital factors influence the number of readmissions for these patients within a specific time fame.

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model = glm(counts ~ patient, data = mydata, familu = poisson(link = „log“))

◊ *Survival analysis:* subjects/objects are followed over time until a terminal event occurs. The hazard function is the probability to „die“ very soon, given that you are „alive“ at time t. The Cox regression model is a proportional hazard regression. It tests the difference between survival groups. Start by creating a survival object from survival time and censoring label; then plot survival curves; finally analyse the Cox regression model. The survival function represents the probability that an event has not occured by time t. The survival curve is a graphical representation if the survival function over time.

In the Cox model, the sign of the coefficients is important. A positive sign indicates thath the hazard (risk of death) is higher, and thus the prognosis worse, for patients with higher value of that variable.

Example: time until progression or relapse of a chronic disease.
prost.surv = Surv(pgtime, event = pgstat, type = „right“)
prost.fit1 = survfit(prost.surv ~ ploidy)
coxmodel = coxph(prost.fit ~ ploidy)

Fixed effects, Random effects, Mixed effects:

◊ *Fixed effects:* describes general behaviour in a popluation, not particular individuals. Fixed effects are estimated using least squares. Fixed effects are constant across individuals.

Example: Suppose you are conduction a study to evaluate the effectiveness of a new medical intervention, such as a drug treatment, on patient outcomes. You have data on multiple patients treated at different hospitals overa a period of time.
model = lm(outcome ~ drug + age) → drug and age are fixed effects.

◊ *Random effects:* describe random draws from population. Random effects are estimated with shrinkage. Random effects vary across individuals. Random effects are used to account for variability in the data that cannot be attributed to observed or measured variables. This variability is assumed to arise from unobserved or unmeasured sources, and it is modeled as a random component. Random effects models are particularly useful in the context of hierarchial or custered data, where observations are not independent but are grouped into clusters or levels.
These models are appropriate when there is variability at different levels that needs to be taken into account. Common scenarios: patients within hospitals, repeated measurements within subjects overtime, repeated measurements on the same subject over time.

Example: Suppose you are conducting a clinical trial to assess the efficacy of a new drug for treating a particular medical condition. You have data from multiple patients who were enrolled in the trial and treated at various medical centers.
model = lmer (outcome ~ (1 | patient)) → patient is random effect

◊ *Mixed effects:* contain both fixed and random effects. These models are particularly useful when dealing with data that exhibit hierarchical or nested structures, such as repeated measurements within individuals, or when there is inherent variability between groups.

Example: Suppose you are conducting a medical study to assess how blood pressure changes over time in a group of patients. You have data from several patients measured at multiple time points, and you want to account for the fact that patients' blood pressure measurements are correlated over time and that there are individual differences between patients.

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```
model = lmer (outcome ~ drug + age + (1 | patient))
```

Mixed-effects model vs. Regression models:

1. *Nested or Hierachical Data:*
    a. *Situation*: When your data has a nested or hierachical structure. A hierarchical structure refers to a system or organization that is arranged in levels or layers, where each level obtains subgroups or elements. The arrangement follows a top-down or nested pattern, with higher levels ecompassing lower levels. In the context of data or statistics, a hierarchical structure often involves nested or grouped data. For example:
        i. *School and students*: Schools can be considered at a higher level, and students within each school form lower-level group. Each school has its own set of patients.
        ii. *Patients in hospital:* Hospitals can be considered at a higher level, and patients within each hospital form lower-level groups. Each hospital has its own set of patients.
        iii. *Repeated measures in experiments:* Time points can be considered at a higher level, and measurements taken at each time point from lower-level groups. Each time point is has its own set of measurements.
    b. *Advantages:* Mixed-effects models allow for the modeling of both fixed effects (population-level effects) and random effects (individual/group-specific effects). This is useful when there is variability at different levels of the hierarchy.

2. *Repeated measures:*
    a. *Situation:* When you have repeated measurements on the same subject over time (longitudinal data) or under different conditions.
    b. *Advantage:* Mixed-effects models naturally handle correlated observations within subject. They account for the correlation structure and allow for the modeling of within-subject variability over time or conditions.

3. *Unbalanced data:*
    a. *Situation:* When the number of observations or measurements per subject or group varies.
    b. *Advantage:* Mixed-effects models can handle unbalanced data more effectively than traditional regression models. The accomodate varying numbers of observations across subjects or groups.

4. *Accounting for individual variation:*
    a. *Situation:* When there is expected heterogeneity between groups or subjects that is not of primary interest but needs to be accounted for.
    b. *Advantage:* Random effects in mixed-effects models allow for modeling individual/group-specific variations, providing a more realistic representation of the underlying population structure.

5. *Increased Statistical Power:*
    a. *Situation:* When there is a variability at different levels (e.g., variability across both subjects and repeated measures).
    b. *Adventage:* By explicitly modeling random effects, mixed-effects models can lead to more efficient estimated and increased statistical power, especially when the design involves multiple levels of variability.

6. *Dealing with missing data:*
    a. *Situation:* When there are missing observations in the dataset.
    b. *Adventage:* Mixed-effects models can handle missing data more gracefully than traditional regression models, provided that the missing data are missing at random or missing completely at random.

Example: Let's consider a hypothetical example involving a study on the impact of a new teaching method on students' test scores. Suppose you have data on test scores from students who were taught using two different methods (Method A and Method B). However, there is a complicating factor: students are nested within

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different schools, and you want to account for the fact that students within the same school may be more similar to each other than to students from other schools

In this example, the test scores of students are influenced by the teaching method (Method A or B), but there is also variability between schools that we want to account for. Each line in the plot represents the test score trajectory for students within a particular school.

The linear regression model (lm_model) treats each observation as independent, ignoring the fact that students are nested within schools. The mixed-effects model (lmer_model) includes a random effect for schools (*(1| SchoolID)),* acknowledging the nested structure of the data.

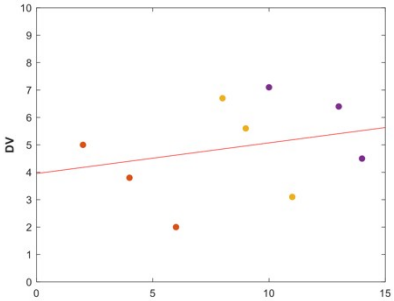
In this example, using a mixed-effects model is essential because it properly accounts for the hierarchical structure of the data. The random effect for schools captures e variability in test scoresbetween schools, providing a more accurate representation of the underlying population structure. The linear regression model, which does not account for the nesting of students within schools, may provide biased estimates and incorrect standard errors.

Cross sectional study design:

Carried out at a one-time point (or short interval) to estimate the prevalence of an outcome of interest. Aims to describe a population, or subgroup, with respect to a set of risk factors. Gives no indication of the sequence of events. Commonly used in public health.

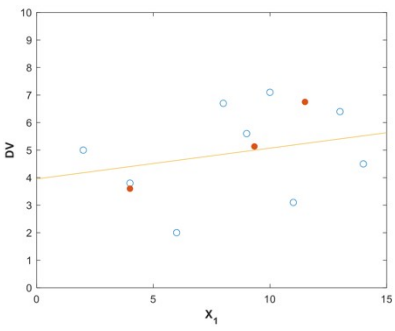
Poolings:

Naive pooling:



The variables need to be independent in order to do a linear regression. In this case, they are not. It is a simple or straightforward method of combining or aggregating data from multiple sources or groups without taking into account the underlying structure or potential differences between the sources or groups.

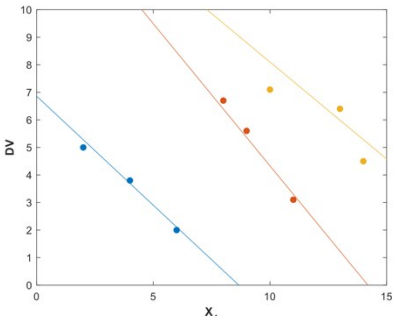
Complete or average pooling:



We are losing features of the dataset. It assumes that there is some shared commonality among the groups while allowing for differences. There is no variability; we are modeling the averages.

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No pooling:



Linear regression model per individual. In the absence of pooling, each group or entity's hierarchical structure which structure is treated as a separate and distinct entity, and there is no sharing of information or parameters between the groups. There is an infinite variability.

Maximum likelihood (ML) and Residual maximum likelihood estimation (REML):

ML: estimate the mean, then estimate the variance of the mean. In small sample sizes, ML is biased in estimating the standard deviation of variance. The primary goal of maximum likelihood estimation is to find the parameter values that maximize the likelihood function, which quantifies how well the model explains the observed data.

REML: variation of normal likelihood. Corrects for bias in the variance component. Less biased in small samples relative to ML. Cannot use in likelihood ratio test to compare models. REML is a statistical method for estimating the parameters of a linear mixed-effects model or a generalized linear mixed-effects model

Bayesian statistics:

Prevalence: Prevalence is a statistical measure that quantifies the proportion or percentage of individuals in a population who have a specific health condition or characteristic at a given point in time or over a specified period.

Incidence: Incidence refers to the rate or frequency at which new cases of a specific disease, condition, or health event occur within a defined population during a given time period.

Predictive values depend on the prevalence of the disease. Sensitivity and specificity don't.

	Pos test	Neg test	Sum
Diseased	TP	FN	**TP+FN**
Healthy	FP	TN	**FP+TN**
Sum	**TP+FP**	**FN+TN**	**Total**

True positive rate: $TPR = sensitivity = \Pr(Test = pos|Truth = pos)$
False positive rate: $FPR = 1 - spec. = \Pr(Test = pos|Truth = neg)$
True negative rate: $TNR = specificity = \Pr(Test = neg|Truth = neg)$
False negative rate: $FNR = 1 - sens. = \Pr(Test = neg|Truth = pos)$

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Positive likelihood ratio:

$$LR+ = \frac{TPR}{FPR} = \frac{TP}{TP+FN} \Big/ \frac{FP}{FP+TN} = \frac{\Pr(Test = pos|Truth = pos)}{\Pr(Test = pos|Truth = neg)} = \frac{sensitivity}{1 - specificity}$$

Negative likelihood ratio:

$$LR- = \frac{TNR}{FNR} = \frac{TN}{FP+TN} \Big/ \frac{FN}{TP+FN} = \frac{\Pr(Test = neg|Truth = neg)}{\Pr(Test = neg|Truth = pos)} = \frac{specificity}{1 - sensitivity}$$

$$Accuracy\ Acc = \frac{TP+TN}{TP+FP+FN+TN}$$

$$Sensitivity = TP/(TP+fiN)$$
$$Specificity = TN/(fiP+TN)$$
$$Pos\ pred\ value = TP/(TP+fiP)$$
$$\neg\ pred\ value = TN/(TP+fiP)$$

Accuracy and precision:

Accuracy: absence of systematic errors. Mean (abs (x – ref))
Precision: absence of random errors. Sd (absolute_error)

Precision can be further specified:

- ◊ *Repeatability:* precision under tight conditions (same equipment, different days). Variation is measured in absolute numbers.
- ◊ *Reproducibility:* precision across labs or devices. Variation is measured in absolute numbers.
- ◊ *Reliability:* how does the variation across measurements relate to the variation in the population? Variation in measurements relative to variation in population.

Confidence intervals for sensitivity/specificity:

	Pos test	Neg test	Sum
Diseased	25	5	**30**
Healthy	15	105	**120**
Sum	**40**	**110**	**150**

Confidence limits for sensitivity:
binom.test(TP, TP + fiN)

Confidence limits for specificity:
binom.test(TN, TN + fiP)

Agreement:

Agreement is characterized by the absence of both systematic and random errors.

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	Pos test A	Neg test A	Sum
Pos test B	20 8.75	5 16.25	25
Neg test B	15 26.25	60 48.75	75
Sum	35	65	100

Agreement: (20 + 60) / 100 = 80% = actual agreement p_0 → *kappa* = 0.529
Ideal agreement: 100% → *kappa* = 1
Worthless agreement p_e = 57.5% (computed from expected frequencies) → *kappa* = 0

Agreement is recaled to *kappa* values using: $k = \frac{p_o - p_e}{1 - p_e}$

Kappa2(mydata)

To calculate whether there is a systematic difference between test A and test B we consider the cases where the tests do not agree (5, 15) and perform *McNemar's* test.
binom.test(c(15, 5)); binom.test(15, 15+5)

	Test A pos	Test A neg
Diseased	25	5
Healthy	15	105

	Test B pos	Test B neg
Diseased	20	10
Healthy	10	110

For sensitivity, we look at the subpopulation of diseased subjects. For specificity, we look at the subpopulation of healthy subjects.

For interval data, weighted kappa can be applied to compensate for „close misses".

Test difference in sensitivity/specificity

Diseased	Test A pos	Test A neg
Test B pos	18	2
Test B neg	7	3

`binom.test(c(2,7))`
gives p=0.180

Healthy	Test A pos	Test A neg
Test B pos	9	1
Test B neg	6	104

`binom.test(c(1,6))`
gives p=0.125

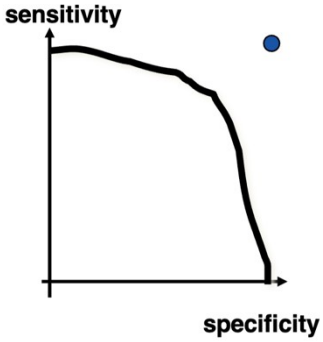
For sensitivity, look at the subpopulation of diseased subjects.

For specificity, look at the subpopulation of healthy subjects.

Threshold levels of a clinical test:

Lower threshold for pathology:
- sensitivity increases
- specificity decreases

Higher threshold for pathology:
- sensitivity decreases
- specificity increases



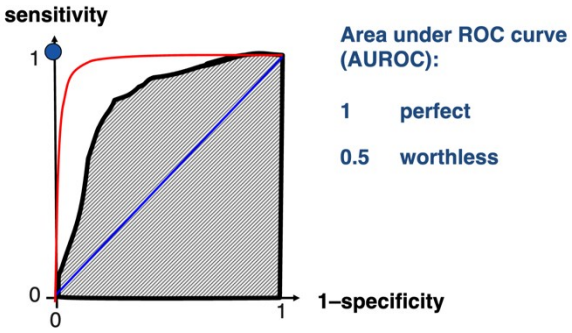
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```
roc.th <- roc (my_data$cancer ~ Tetrahydrocortisone, data = my_data)
roc.pr <- roc (my_data$cancer ~ Pregnanetriol, data = my_data)
plot (roc.th, col = "blue", asp = 1, main = "Diagnosing adrenal cancer")
lines (roc.pr, col = "red")

roc.th$specificities
roc.th$sensitivities
roc.th$thresholds
(roc.th$specificities + roc.th$sensitivities)
max.index <- which.max(roc.th$specificities+roc.th$sensitivities)
(roc.th$specificities+roc.th$sensitivities)[max.index]

roc.th$thresholds[max.index]
roc.th$specificities[max.index]
roc.th$sensitivities[max.index]
```

ROC curve:



ROC curve is a graphical tool used in statistics and machine learning to evaluate the performance of binary classification models where the distinction between two classes is important. The ROC curve is a useful way to visualize and assess the trade-off between the true positive rate (sensitivity) and the false positive rate (1-specificity) at various classification thresholds.

A greater area under the curve means a better prediction. To compare two ROC curves a *DeLong test* is implemented.

```
roc.test (roc.th, roc.pr)
```

Bland-Altman plot:

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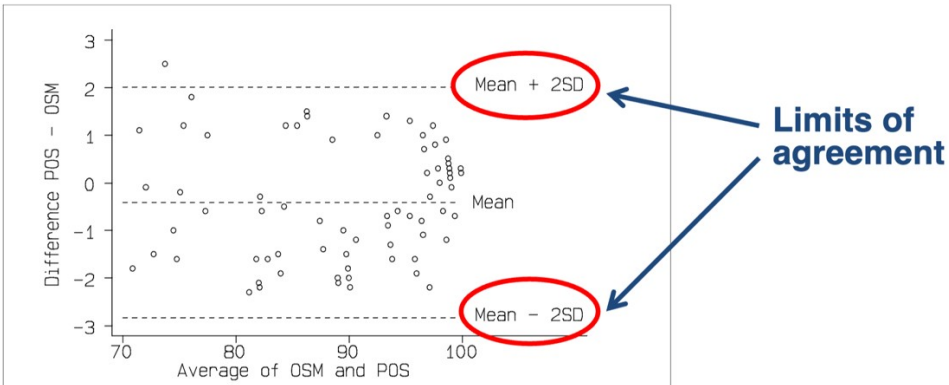


Fig 3. Oxygen saturation monitor and pulsed saturation oximeter

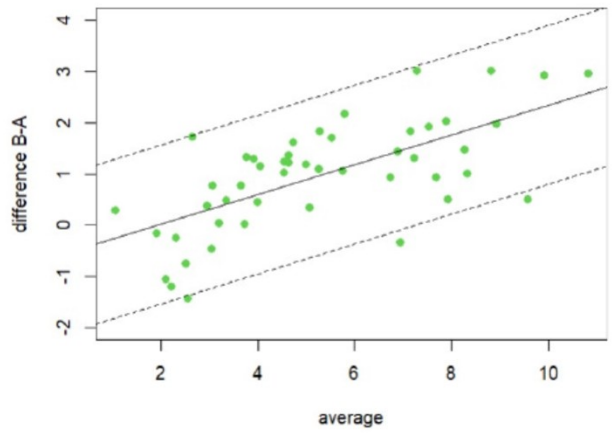
From this plot, it is possible to deduce a lot of different features:

1. **Mean Bias (Bias Line):** The horizontal line in the Bland-Altman Plot represents the mean bias or the mean of the differences between the two measurement methods. If this line is close to zero, it suggests that, on average, there is little or no bias between the methods. If it deviates significantly from zero, it indicates a systematic difference in measurements.
2. **Precision (Spread of Data):** The spread of data points around the mean bias line gives you an idea of the precision of the two methods. If the data points are tightly clustered around the mean bias line, it suggests good precision and agreement between the methods. Wider dispersion implies lower precision. Precision must be utilized if it is assessed the agreement between a reference and the measurements provided by a device. *It doesn't make sense to evaluate the precision by looking at the values produced by two devices that are measuring the same object.*
3. **Limits of Agreement (95% Confidence Intervals):** The two dashed lines in the Bland-Altman Plot represent the 95% confidence intervals for the differences between the methods. These lines provide a range within which approximately 95% of the differences are expected to fall. Wider limits of agreement indicate greater variability in the measurements

Example: comparing glucose levels taken from patients at two different hospitals. How much do these two hospitals "agree"?

```
average  <-  (HospA + HospB) / 2
difference  <-  HospB - HospA
mean(difference) + 1.96*sd(difference)
mean(difference) - 1.96*sd(difference)

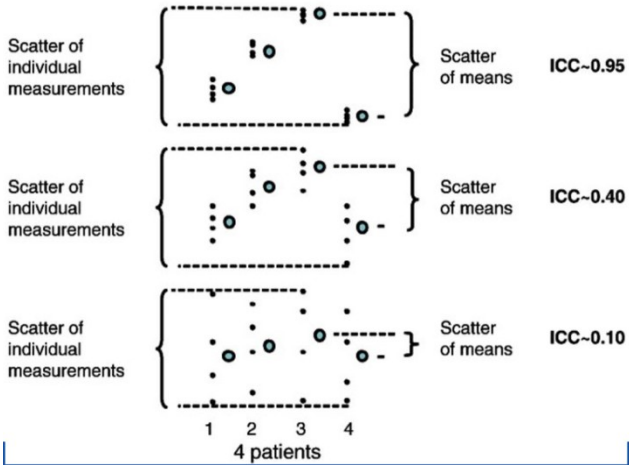
plot(average, difference, ylim = c(-2,4), pch = 16, col = 4, ylab = "difference B - A")
abline(h =  mean(difference))
abline(h = mean(difference) + 1.96*sd(difference), lty = 2)
abline(h = mean(difference) - 1.96*sd(difference), lty = 2)
```



This is an alternative solution. We are accounting for a linear trend. The concept is always the same: we expect 95% of observations to fall in between the dashed lines. This time, however, the bands are narrower.

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Intraclass correlation coefficient (ICC):



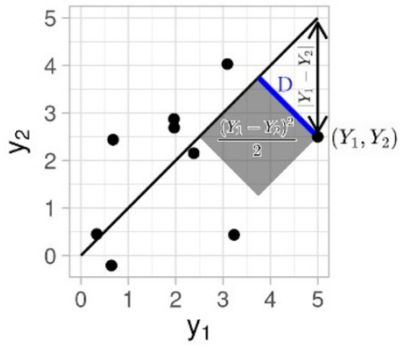
Compares the spread of individual measurements to the spread of means for each measurement object (subject). It is a statistical measure used to assess the reliability and consistency of measurements or ratings made by different observers or devices on the same subjects or objects. It is particularly useful in situations where you want to quantify the degree of agreement or consistency between multiple measurements or raters.

The higher the ICC value, the higher the reliability. Data needs to be continuous.

Can only handle cases with one measurement by each rater for each case.

If all the cases are judged by the same raters, we use a “two-way model”. If each case is judged by a different set of raters, we use a “one-way model”.

Example: 10 patients with radius fracture, angulation of the fragment measured by 10 radiologists (raters)
icc(ratings, model = "two", type = "agreement", conf.level = 0.95)
CCC(radiusdata$Rater_A, radiusdata$Rater_B, conf.level = 0.95)$rho



An alternative to ICC is Lin‘s concordance correlation coefficient (CCC). It is an estimate of the squared distance to the identity line. Numerically, it is very close to ICC but it is less general.

A measure of distance from the 45-degree line naturally quantifies the (dis)agreement between the two sets of observation.

ICC and CCC are used for continous interval data, kappa fro categorical data.

Kappa value:

Kappa (κ) is a statistical measure that assesses the agreement between two raters or observers who classify items into categories. It is commonly used in inter-rater reliability studies when there are categorical ratings. The kappa statistic considers the agreement that could occur by chance and then measures the actual agreement beyond chance.

The kappa statistic ranges from -1 to 1:

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- κ=1 indicates perfect agreement.
- κ=0 suggests agreement equal to chance.
- κ<0 suggests agreement is worse than chance.

Here's a more intuitive interpretation of kappa values:

- 0.81 − 1.0 0.81−1.0: Almost perfect agreement
- 0.61 − 0.80 0.61−0.80: Substantial agreement
- 0.41 − 0.60 0.41−0.60: Moderate agreement
- 0.21 − 0.40 0.21−0.40: Fair agreement
- 0.01 – 0.20 0.01−0.20: Slight agreement
- < 0: Poor agreement (worse than chance)

Example: Let's consider a scenario in medical research where doctors are assessing the presence or absence of a medical condition in patients, such as the detection of a specific symptom or disease. Imagine you have a dataset where two radiologists are independently examining medical images (e.g., X-rays, MRIs) of patients for the presence or absence of a particular condition (e.g., the presence of a tumor). Now, you want to assess the agreement between the two radiologists in diagnosing the presence or absence of the medical condition.

Organising data:

- ◊ Each variable must have its own column.
- ◊ Each observation must have its own row.
- ◊ Each value must have its own cell.
- ◊ Never mix different variables in one column.
- ◊ Never put data from different cases in the same row.
- ◊ Kepp data from one case in one row.

Case	Rater	Angle
1	Rater_A	19,5
2	Rater_A	18,1
3	Rater_A	31,0
4	Rater_A	16,7
5	Rater_A	22,7
6	Rater_A	49,3
7	Rater_A	30,6
8	Rater_A	14,4
9	Rater_A	23,5
...	...	...
8	Rater_D	19,1
9	Rater_D	27,4
10	Rater_D	15,9

Case	Rater A	Rater B	Rater C	Rater D
1	19,5	24,2	15,3	22,7
2	18,1	14,1	12,4	11,2
3	31,0	32,1	31,8	31,6
4	16,7	15,5	13,3	20,5
5	22,7	22,8	26,9	24,8
6	49,3	44,5	43,4	46,0
7	30,6	24,5	22,0	30,7
8	14,4	15,4	17,3	19,1
9	23,5	26,6	22,0	27,4
10	11,8	13,8	19,4	15,9

Data can be organized in a wide or long form.

For complex statistical models (mixed effect models) or ICC calculations, long form is usually preferred. Data can be reshaped from long form to wide form and vice versa.

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From wide to long format: gather(table, key, cases, y1:y4); pivot_loger(table, cols = y1:y4, names_to = „key“, value_to = „cases“)

From long to wide format: spread(table, key, value); pivot_wider(table, names_from = key, values_from = value)

Confidence intervals:

Use a known estimate of the parameter θ and of the standard error (SE).

To get a 95% confidence interval: [θ – 1.96 SE, θ + 1.96 SE]

Alternative methods for obtaining interval estimates, standard errors and confidence intervals are:

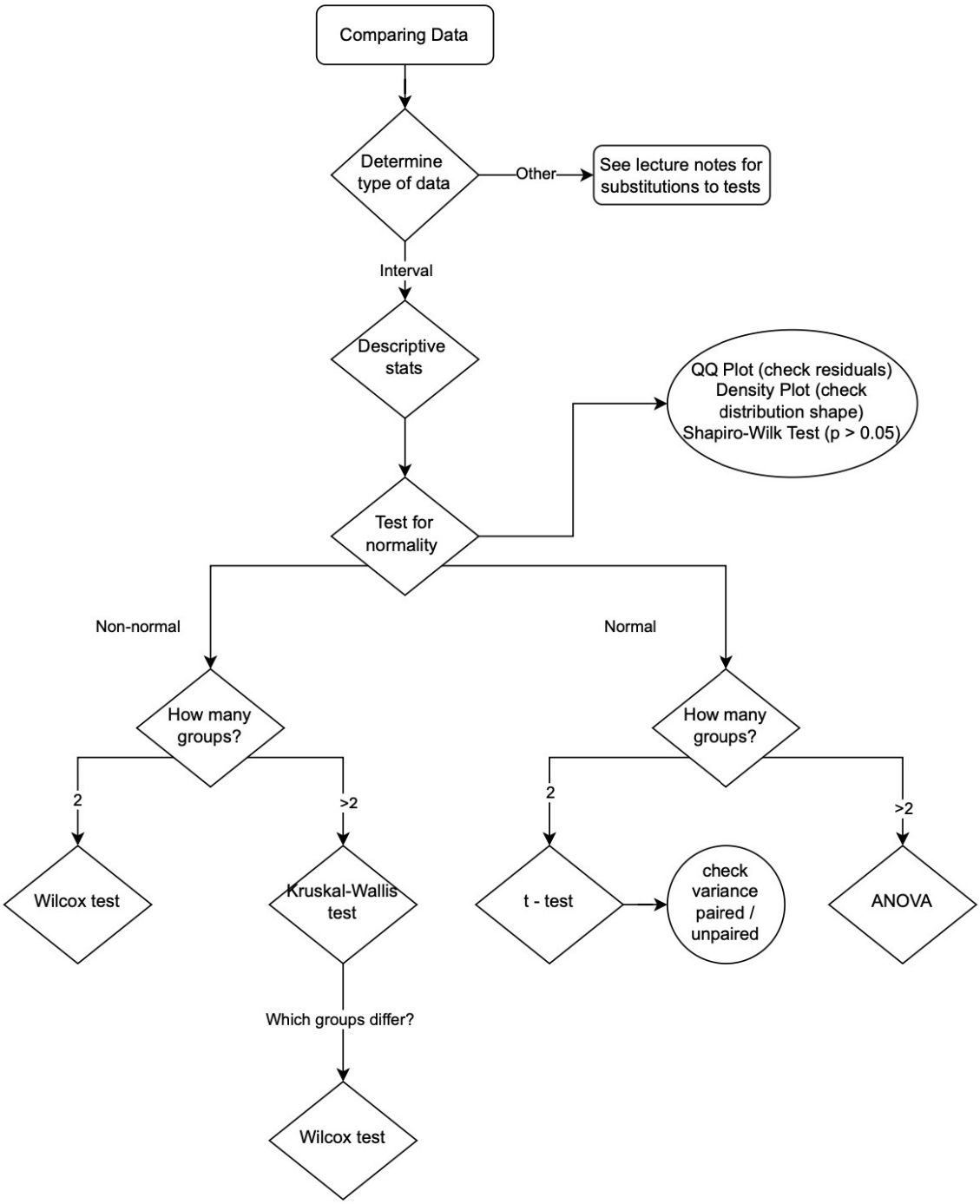
- ◊ **Jack-knife:** early approach to resampling. This method recomputes estimates of a statistic with smaller and smaller subsets of the original sample. It is a deterministic process (same result each time). Gives an estimate of the SE of, for example, an estimate of the mean.

- ◊ **Bootstrapping:** more computationally demanding. It is more flexible. It is a non-deterministic process (slightly different results each time). This method does random sampling with replacement from the original sample. It is dependent on the number of samples.
  Example: estimating the mean by resampling.

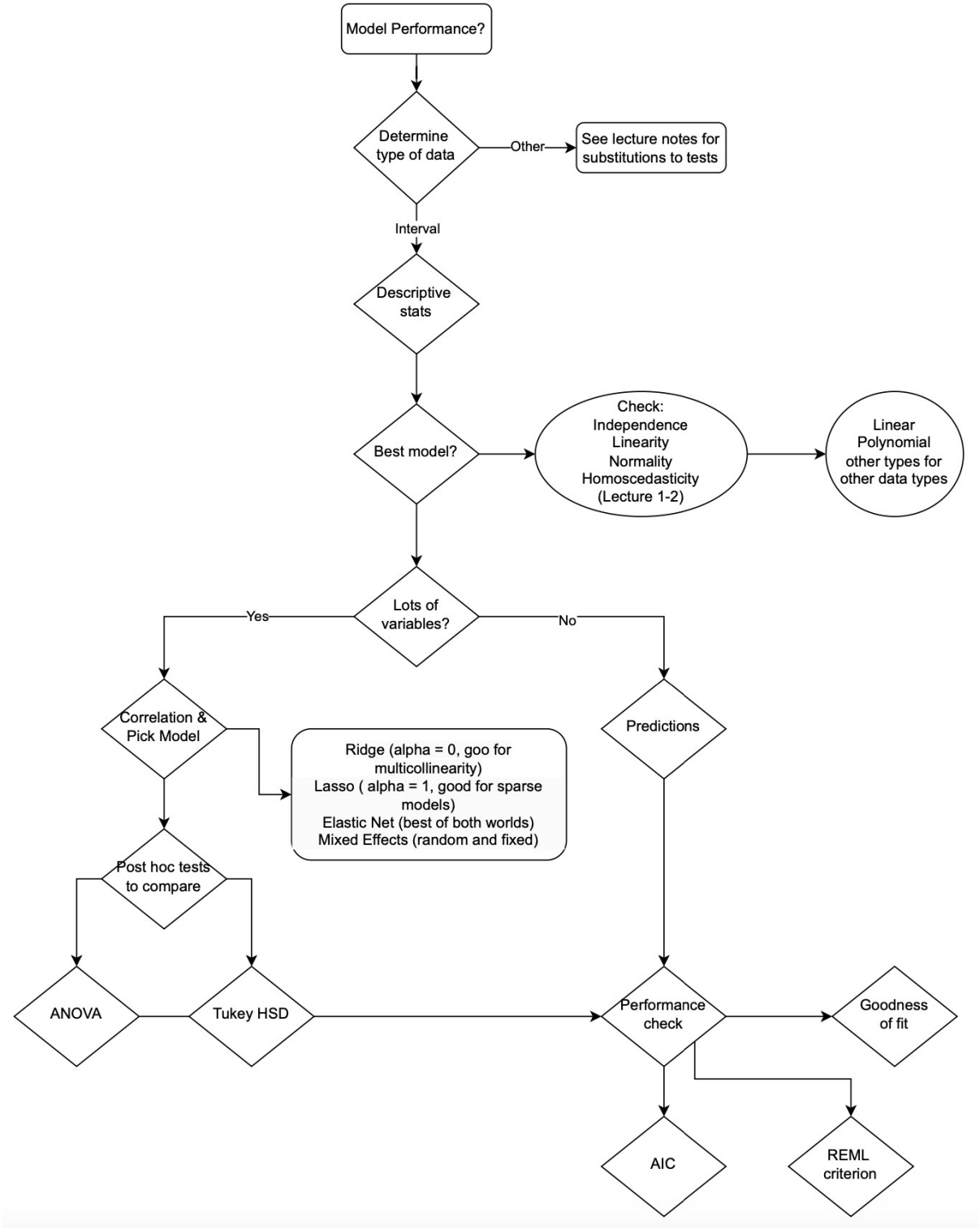
- ◊ **Permutation tests:** non-parametric tests applicable to any distribution. The basic idea is to go through all possible rearrangements of the observed data. Requires heavy computations, especially with large datasets.
  Example: study the probability of death with a new treatment, depending on sex, age, diagnosis and treatment type.

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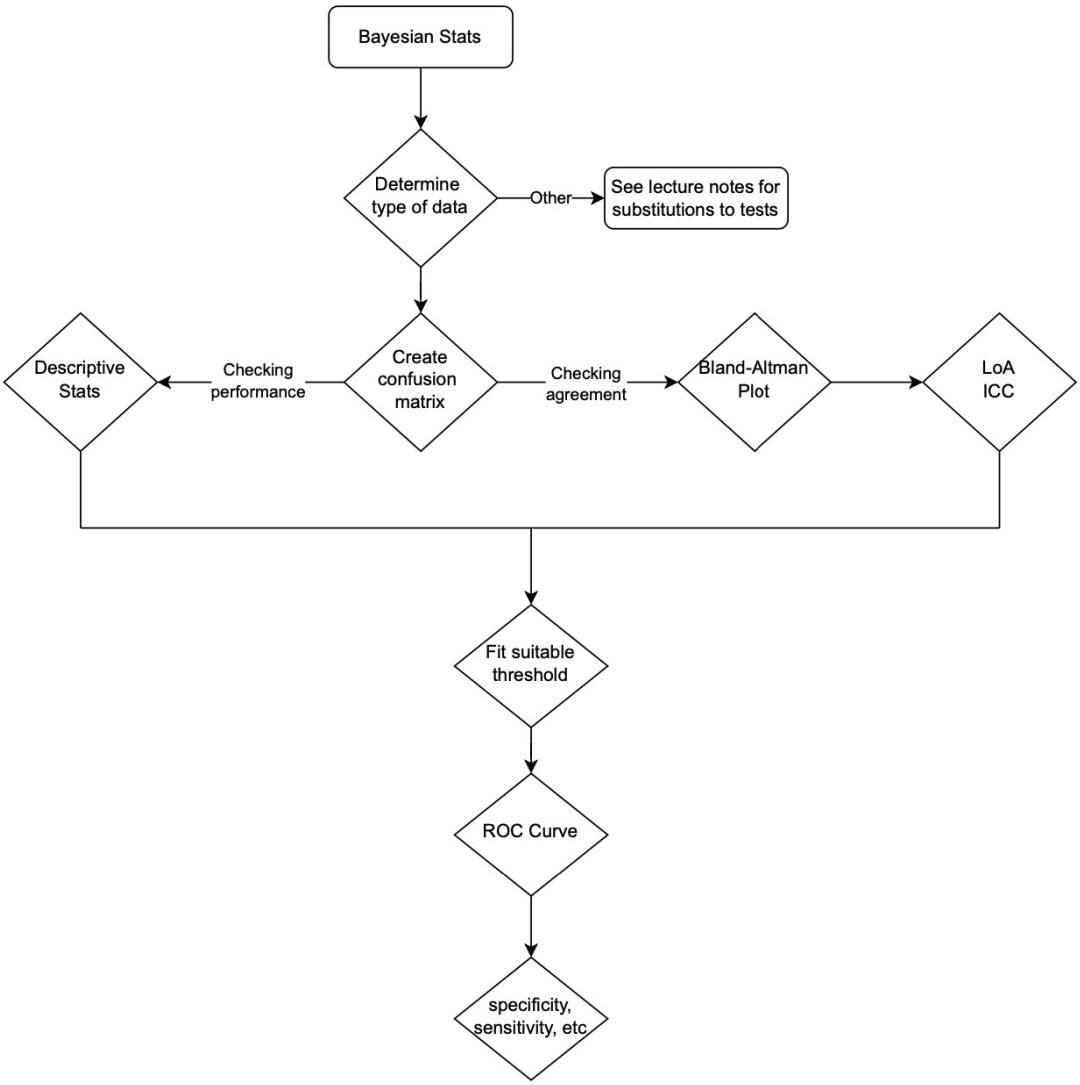
Summary:



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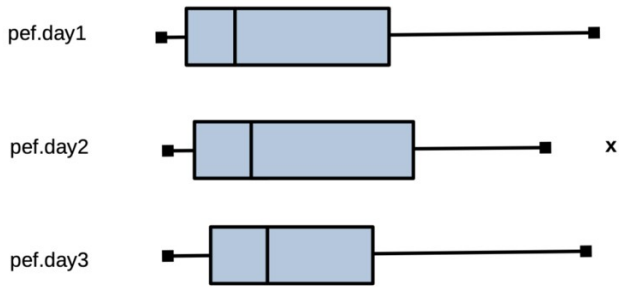
Final exam 231205:

Question 1:

A pulmonologist (medical specialist in lung diseases) wants to study how an important measure of lung function, the peak expiratory flow (PEF; in l/min), varies between days, in response to weather changes. Three days in a row, she makes measurements of PEF in 20 patients, and then she summarises the results in a table like this:

id	pef.day1	pef.day2	pef.day3
1	367	399	350
2	411	454	438
3	352	395	337
...			
20	404	447	422

The distribution of values is illustrated by the following box plot:



To find out whether there is a systematic difference in PEF between these three days, which statistical method would you recommend? Please motivate your answer carefully. (4 p)

Answer:

- *The data are on an interval scal.*
  - *Reason: The PEF measurements are continuous quantities that have meaningful differences between values but lack a true zero point.*
  - *For example, a difference of 50 l/min (e.g., 400 - 350) has the same meaning regardless of the absolute values, which is a hallmark of an interval scale.*
  - *The absence of a true zero (a point where no lung function exists) prevents the data from being on a ratio scale. If it were a ratio scale, statements like "200 l/min is half of 400 l/min" would be valid.*

- *Since the same individuals were measured on repeated occasions, the analysis needs to compare variables (columns) rather than groups (rows), so a test for related samples must be used.*
  - *Reason: The same 20 individuals were measured on three separate days (repeated measures), meaning the data are dependent. Each individual serves as their own control, which requires a statistical method designed for related samples.*
    - *Implication: Tests that assume independence between samples (e.g., one-way ANOVA) are not suitable here. Instead, a test for related samples is necessary.*

- *As the boxplot suggests that the variables are not normally distributed (they are right-skewed), a non-parametric test is preferable.*

- *A non-parametric test for three or more related interval variables is the Friedman test, which is suitable in this situation.*

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- *It does not assume normality, making it ideal for this situation where the data appear skewed.*
- *It is specifically designed for repeated measures, addressing the dependence between observations.*

Question 2:

A team of abdominal surgeons are evaluating a new chemical biomarker (blood test) for cancer of the large bowel. After 426 patients had been examined with the new test and the clinical follow-up results had been obtained, the following results were obtained:

test result	final diagnosis	n
pos	pos	65
pos	neg	19
neg	pos	8
neg	neg	334

a) Estimate the sensitivity and specificity of the test while taking into account the uncertainty of your calculations (2 p)
b) Is there evidence of systematic over- or underestimation of disease by the test? Motivate your answer. (1 p)
c) Given that a patient from this population has received a positive outcome of the blood test, what is the probability that he has cancer of the large bowel? Again, take the statistical uncertainty into account. (1 p)
d) Given that a patient from this population has received a negative outcome of the blood test, what is the probability that he does not have cancer of the large bowel? Again, take the statistical uncertainty into account. (1 p)

Answer:

a) We have tp = 65, fp = 19, fn = 8, and tn = 334.

- *Calculate sensitivity:*
  binom.test(tp, tp + fn)
  gives a sensitivity of 0.890 with 95% confidence limits 0.795 and 0.951.
- *Calculate specificity:*
  binom.test(tn, fp + tn)
  gives a specificity of 0.946 with 95% confidence limits 0.917 and 0.967.

Sensitivity *measures the proportion of actual positive cases (cancer patients) correctly identified by the test.* $Sensitivity = \frac{TP}{TP+FN}$

Specificity *measures the proportion of actual negative cases correctly identified by the test.*

$$Specificity = \frac{TN}{TN+FP}$$

b) To evaluate over- or underestimation of a binary variable (i.e., to test whether there is a significant difference between two related binary columns), we can use the McNemar test, comparing false positives to false negatives:

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binom.test(fp, fp + fn)
which gives a p-value = 0.05224. Thus, there is no significant systematic difference between the test and the reference method. In other words: the test does not significantly over- or underestimate the disease.

Why McNemar's Test is used: McNemar's test evaluates whether there is a significant difference between two related proportions. In this case, it compares ***false positive and false negative****.*
If FP≠FN, the test is systematically over- or underestimates disease.

c) *This probability is called the positive predictive value (PPV). It is obtained by:*
binom.test(tp, tp + fp)
which gives PPV=0.774 with 95% confidence limits 0.670 and 0.858.

d) *This probability is called the negative predictive value (NPV). It is obtained by:*
binom.test(tn, tn + fn)
which gives NPV=0.977 with 95% confidence limits 0.954 and 0.990.

Question 3:

A group of 39 patients with Covid infection were followed for 6 months after the onset of disease. It turned out that 27 of the patients had recovered within 6 months, whereas 12 had not. The data are found in the file recov.xlsx.

Try to use these data to determine whether – and in what direction – age and sex influence the probability of recovery after Covid infection. (4 p)

Answer:

Since the variable to be predicted (recovery) is binary, we should use binary logistic regression with recovery as dependent variable, and age and sex as predictors.

This is done with the following R code:

```
library(readxl)
recov <- read_excel("Downloads/recov.xlsx")
recov

logistic.model <- glm(recovery ~ age + sex, family=binomial, data = recov)
summary(logistic.model)
```

```
Call:
glm(formula = recovery ~ age + sex, family = binomial, data = recov)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  12.9091     4.3006   3.002  0.00268 **
age          -0.2417     0.0827  -2.923  0.00347 **
sexM          1.6607     1.0943   1.518  0.12910
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 48.145  on 38  degrees of freedom
Residual deviance: 28.517  on 36  degrees of freedom
AIC: 34.517

Number of Fisher Scoring iterations: 6
```

Since the p value for age is significant (0.00347) *and the corresponding regression coefficient is negative, the probability of recovery decreases with age.*

Since the p value for sex is non-significant (0.12910)*, there is no evidence that sex influences the probability of recovery after Covid.*

Question 4:

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A researcher in medical care is working on improving a questionnaire for assessing the health status and well-being of patients. The questionnaire consists of a number of questions which are scored and then summarised in an evaluation of the patient's health status with the following labels:

A – excellent health status
B – good health status
C – normal health status
D – poor health status
E – very poor health status

As the researcher suspects that the precision of the questionnaire is insufficient, he wants to test a new version of it and compare that to the old version. Two weeks in a row, 100 patients complete both the old version and the new version, and the results are gathered in a table like this:

oldversion.week1	oldversion.week2	newversion.week1	newversion.week2
B	B	B	B
A	B	A	A
C	C	D	D
D	D	D	E
A	A	A	A
...	...	...	...

The researcher then performs the following calculations in R:

```
> col5 <- (oldversion.week1==oldversion.week2)
> col6 <- (newversion.week1==newversion.week2)
> col7 <- (col6>col5)
> col8 <- (col6==col5)
> col9 <- (col6<col5)
> binom.test(c(sum(col7), sum(col9)))

      Exact binomial test

data:  c(sum(col7), sum(col9))
number of successes = 12, number of trials = 15, p-value = 0.03516
alternative hypothesis: true probability of success is not equal to 0.5
95 percent confidence interval:
 0.5191089 0.9566880
sample estimates:
probability of success
                   0.8
```

Looking at the results, he draws the conclusion that the new version has better precision than the old version.

a) Is this conclusion correct? Please motivate your answer, while paying particular attention to the terminology. (2 p)
b) What is the name of the statistical test performed in this context? (1 p)
c) Can you suggest better names for the variables col5-col9 ? (1 p)
d) To make the test more fine-tuned, could you suggest a modification of the statistical analysis? (1 p)

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Final exam 221205:

Question 1:

Elevated concentrations of Prostate-specific antigen (PSA) may indicate prostatic cancer. A hospital laboratory for clinical chemistry is considering how to choose the limit for pathological levels of PSA. They have for a long time applied a limit of 3 μg/l, but are now asking themselves whether a different threshold value would be more appropriate. They have accessvto a material of 200 patients in which PSA was measured and a histopathological diagnosis of cancer or no cancer is available.

Suggest a method for selecting an optimal threshold level for diagnosing prostatic cancer. This should include a suitable statistical tool, and a general principle for interpreting the outcome of that tool. Also indicate what additional considerations must be included in this process. (3 p)

Answer:

- *A tool very well suited for situations where a classification threshold is to be set is Receiver operating characteristic (ROC) analysis.*
- *Each point on the ROC curve represents a threshold level. Points close to the upper left corner are better.*
- *The selection process must also take into account the relative importance (or "cost") of false negatives (missed cases with cancer) and false positives (cases wrongly suspected of having cancer).*

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Question 2:

In a clinical study of anti-hypertension medication (drugs against high blood pressure), two drugs are compared (A and Placebo) in 28 patients. The trial was a cross-over study, so that each patient received drug A during one period of 4 weeks and Placebo during one period of 4 weeks. In 14 patients, the drug A period came before the Placebo period, and in 14 patients it came after. Both patients and clinical doctors were unaware of which treatment was given when.

Measurements of systolic blood pressure at the end of each treatment period in the 28 patients are found in the file btdata.xlsx.

a) Discuss what statistical method should be used to compare the measurements, perform the required calculations, and formulate a conclusion from the results. (3 p)

b) Suggest a graphical presentation method for the data. Motivate your choice of graph. (2 p)

c) Discuss how the data analysis could have been made better if more information had been available in the data file. Please mention both which data to add and which statistical technique to use. (2 p)

Answer:

a) Since the data are paired (two measurements for each patient), a paired test must be used.
A look at the histograms, or a test of normality such as Shapiro-Wilk, (in both cases most relevant for the difference between Drug A and Placebo) shows no indication of significant deviation from normality. Therefore, a parametric test can be used.

```
    ...1               bpA          bpPlacebo
 Length:28          Min.   :110.0   Min.   :148.0
 Class :character   1st Qu.:152.8   1st Qu.:164.0
 Mode  :character   Median :165.0   Median :170.0
                    Mean   :163.8   Mean   :172.6
                    3rd Qu.:174.0   3rd Qu.:180.0
                    Max.   :197.0   Max.   :204.0
> shapiro.test(test_A) # data normally dist

        Shapiro-Wilk normality test

data:  test_A
W = 0.9645, p-value = 0.4431

> shapiro.test(test_placebo) # data normally dist

        Shapiro-Wilk normality test

data:  test_placebo
W = 0.96578, p-value = 0.4732
```

```
> t.test(test_A, test_placebo, paired = TRUE)

        Paired t-test

data:  test_A and test_placebo
t = -3.2193, df = 27, p-value = 0.003335
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -14.385383  -3.186046
sample estimates:
mean difference
      -8.785714
```

Thus, there is a significant difference between the treatments, with lower blood pressure for Drug A (mean 164) than for Placebo (173).

b) A boxplot shows that the the blood pressure is lower for Drug A (median 165) than for Placebo (median 170).

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Question 3:

In a study of image quality in nuclear medicine imaging (bone scintigraphy), a clinician was asked to rate the visibility of two anatomical structures (the left clavicle and the right femoral condyle) in images acquired with two different cameras (Old and New), using two different doses of radioactive technetium (Tc99m) (500 and 1000 Mbq). The visibility was rated according to the scale:

(1) not visible
(2) just perceivable
(3) diagnostically acceptable
(4) well defined

The data are summarised in a table:

id	scanner	dose	clavicle	femur
1	Old	500	1	2
2	Old	1000	3	3
3	Old	500	1	1
4	Old	1000	3	1
5	Old	500	1	1
6	Old	1000	1	2
...	...	...	...	...
119	New	500	1	2
120	New	1000	2	4

Analysis was performed in R with the following results:

```
> table(scanner, clavicle)
       clavicle
scanner  1  2  3  4
    Old 39 11 10  0
    New 19 24  8  9
> table(scanner, femur)
       femur
scanner  1  2  3  4
    Old 21 27  5  7
    New 23 24  5  8
> table (dose, clavicle)
      clavicle
dose    1  2  3  4
  500  41 19  0  0
  1000 17 16 18  9
> table (dose, femur)
      femur
dose    1  2  3  4
  500  26 34  0  0
  1000 18 17 10 15
>
> cmodel <- polr(as.factor(clavicle)~scanner + dose, Hess=TRUE)
> summary(cmodel)
Call:
polr(formula = as.factor(clavicle) ~ scanner + dose, Hess = TRUE)

Coefficients:
             Value Std. Error t value
scannerNew 1.954692  0.4276003   4.571
```

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```
dose       0.005413  0.0009284   5.830

Intercepts:
    Value  Std. Error t value
1|2 4.8449 0.8245     5.8763
2|3 6.6876 0.9343     7.1576
3|4 8.3892 1.0655     7.8732

Residual Deviance: 227.7131
AIC: 237.7131
> tidy(cmodel, p.values=TRUE)
# A tibble: 5 × 6
  term       estimate std.error statistic   p.value coef.type
  <chr>         <dbl>     <dbl>     <dbl>     <dbl> <chr>
1 scannerNew  1.95     0.428         4.57  7.74e- 7 coefficient
2 dose        0.00541  0.000928      5.83  1.84e-11 coefficient
3 1|2         4.84     0.824         5.88 NA        scale
4 2|3         6.69     0.934         7.16 NA        scale
5 3|4         8.39     1.07          7.87 NA        scale
> brant(cmodel)
------------------------------------------
Test for    X2    df    probability
------------------------------------------
Omnibus           1.85  4     0.76
scannerNew  1.85  2     0.4
dose        0     2     1
------------------------------------------

H0: Parallel Regression Assumption holds
...

> fmodel <- polr(as.factor(femur)~scanner + dose, Hess=TRUE)
> summary(fmodel)
Call:
polr(formula = as.factor(femur) ~ scanner + dose, Hess = TRUE)

Coefficients:
              Value Std. Error t value
scannerNew -0.075733  0.3434130 -0.2205
dose        0.002758  0.0007438  3.7079

Intercepts:
    Value   Std. Error t value
1|2  1.3285  0.5632     2.3590
2|3  3.4156  0.6471     5.2782
3|4  4.1082  0.6899     5.9548

Residual Deviance: 272.6487
AIC: 282.6487
> tidy(fmodel, p.values=TRUE)
# A tibble: 5 × 6
  term       estimate std.error statistic   p.value coef.type
  <chr>         <dbl>     <dbl>     <dbl>     <dbl> <chr>
1 scannerNew -0.0757   0.343      -0.221   0.825    coefficient
2 dose        0.00276  0.000744    3.71    0.000109 coefficient
3 1|2         1.33     0.563       2.36   NA        scale
4 2|3         3.42     0.647       5.28   NA        scale
5 3|4         4.11     0.690       5.95   NA        scale
> brant(fmodel)
------------------------------------------
Test for    X2    df    probability
```

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```
--------------------------------------------
Omnibus            0.34  4     0.99
scannerNew  0.34   2     0.85
dose        0      2     1
--------------------------------------------
H0: Parallel Regression Assumption holds
```

What conclusion can be drawn from the output? (4 p)

Answer:

- *The use of ordinal logistic regression is appropriated since the dependent variables are defined on an ordinal scale, and the Brant test does not indicate a violation of the parallel regression assumption.*

- *For the Clavicle, the image ratio quality is significantly higher with the new scanner than with the old scanner (p-value < 1e-8). For the Femur, there is no such difference (p-value = 0.825).*

- *For both bones, the image quality is significantly higher with a higher dose of technetium (p-value < 1e-9 and p-value < 0.000109, respectively).*

- *The AIC score for both models is low which is good, meaning that the models fits the data well.*

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Question 4:

A service engineer, Peter, provides technical support to CT scanners, including replacement of the X-ray tube, either because the tube has ceased to function or because a date specified in the customer contract has been reached. His activities concerning a certain type of scanner are summarised in the table below. There is also data on the number of exams per month typically carried out at each site.

tube number	installation	removal	cause	exams
0001	2019-02-14	2021-04-14	malfunction	500
0002	2019-02-21	2019-06-29	malfunction	450
0003	2019-02-25	2020-01-14	contract	200
0004	2019-03-02	2020-06-30	contract	155
0005	2019-03-15	2021-02-04	contract	230
0006	2019-03-20	2020-08-14	malfunction	650
...	...	...	...	...
0049	2021-12-10	2022-08-13	contract	420
0050	2021-12-20	2022-10-02	contract	275

Peter suspects that the time until the tube ceases to function is shorter at sites where the scanner is used for a larger number of exams each month. How should such a hypothesis be tested? Please motivate your answer, and specify both the general type of analysis and a statistical test or model that would be useful. Also, try to formulate one or two lines of R code to be used in the analysis (5 p)

Answer:

- *We are studying the time until a terminal event occurs in some, but not all, cases.*
- *This makes the survival analysis the preferred method.*
- *To test the hypothesis one may use Cox proportional hazard regression, with exams as predictor. [Alternatively, the log-rank test after splitting the data in two groups by number of exams].*

For example, the follwing code:

```
tube.serv = Surv(time = (removal ~ installation), event = (cause == „malfunction“))
coxph(tube.surv ~ exams)
```

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Question 5:

A medical researcher, Sara, uses a large cross-sectional population database with more than 7,000 cases to study how serum albumin varies with age and sex. She has found that albumin decreases with age, and now wants to get deeper insight in this finding.

The data are found in albumindata.xlsx.

a) For answering whether the decrease with age is more pronounced in women or in men, which of the following R models would be best?

   What conclusion should Sara draw?

   ```
   model1 <- lm(albumin ~ sex)
   model2 <- lm(albumin ~ age + sex)
   model3 <- lm(albumin ~ age*sex)
   model4 <- lmer(albumin ~ age +(1|sex))
   ```

   Please motivate briefly (in one or two sentences) why each of the models is more or less appropriate. (4 p)

b) For answering whether the decrease occurs faster or slower in high age, which of the following R models would be best?

   What conclusion should Sara draw?

   ```
   model5 <- lm(albumin ~ log(age))
   model6 <- lm(albumin ~ exp(age))
   model7 <- lm(albumin ~ age + age^2)
   model8 <- lm(albumin ~ poly(age,2))
   ```

   Again motivate briefly why each of them is more or less appropriate. (4 p)

Answer:

A)

- *Model 1 only considers sex differences, not any dependence on age, so it is clearly not suitable for answering the question.*
- *Model 2 studies how albumin varies with age and with sex, but not whether the age dependence differs between· men and women.*
- *In model 3 (which may also be written as albumin ~ age + sex + age:sex), the interaction term age:sex describes exactly how the age decline differs between the sexes.*
- *Model 4 is a mixed-effects model treating sex as a random effect, which is not reasonable. (There is no larger population of sexes of which male and female are a sample.)*
- *Thus, model 3 is the best choice.*
- *In the output of model 3, a significant p value for age:sexMale (p<2e-16) and a negative coefficient (-0.0065851) indicate that the difference present between males and females (with men having higher values) becomes smaller with age.*

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(B)

- *Model 5 can describe a gradually changing slope in the age dependence, but there is not one specific term related to the change in slope, just one parameter describing both the slope and its change over time.*

- *The same is true for model 6*

- *Model 7 looks like a quadratic model where the change in slope with age is described by a second-order term, but due to how R is constructed, it will not work, and you will only get one parameter related to age.*

- *Model 8 is the proper way of defining a quadratic model in R. You will get one parameter for the linear dependence and one for the second-order dependence on age, and the latter term will describe just how the slope changes over time.*

- *Therefore, model 8 is the best choice.*

- *In the output of model 8, the non-significant p value for poly(age, 2)2 (p=0.912) indicates that there is no change in slope over the studied age span.*

  (The data is a subset of the dataset nhanes2 in the MASS package in R.)

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